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# Rust courseproject report

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## Shell Script

Made a custom shell in Rust supporting 9 commands: ls, cd, pwd, write, cat, rm, du, help, exit. The shell includes flag parsing, input tokenization with quoting and escaping support, and consistent error handling across the commands.

I developed and tested in WSL (Windows 11, Ubuntu 22.04.02) using VS Code.

External dependencies (only 1): Chrono for formatting file modification timestamps

I used the standard library crates like std::io, std::fs, std::path, std::env, and also std::os::unix::fs for the permissions and disk block sizes.

With this project I've learned to better understand and work with the ownership/borrow model, module system, trait derivation, iterators, and handling of Result/Option types. I've also learned a lot about what kind of code there is behind a shell, and how permissions, blocks, and timestamps are handled (at least in Linux/Unix).

## Features done

*List features here*

- *Required features*
- *cd supports both relative and absolute paths, as well as ~ for home directory*
- *ls -a to show hidden files in addition to normal output*
- *rm with support for the -r flag to delete files and directories*
- *du with flags -a, -b, -h, and -c to show disk usage of files*
- *cat to display one or more files*
- *Input and output support white spaces in paths and filenames via quotation and escaping*
- *exit in order to exit the program*
- *All commands support -h for help (except in du it's –-help), can also be used as help <command>*

*An enum (CmdResult) was used to keep the control flow in main(), circumventing the need for commands to call std::process::exit.*

*All commands resolve relative paths based on the shell's tracked working directory instead of the process's actual cwd, allowing commands after cd to behave as desired.*

*display_arg wraps filenames containing spaces in quotes in output, making clear which words are part of the same name*

*Filesystem errors are caught and printed to stderr with the command name prefix, avoiding panicking*

## Link to video

*https://drive.google.com/file/d/1g1j0dDYZNB08A7uAZFFEiB1-ohhCeDvw/view?usp=sharing*